

BankSmart

Customer Spending Behavior Analysis for Improving Savings & Reducing Churn

Problem Statement

In the modern digital banking environment, customer retention and engagement are critical challenges. A mid-sized retail bank is experiencing slow growth in savings accounts, high churn rates among customers, and underutilization of digital banking channels. The bank seeks to understand customer spending behavior and financial patterns to identify key factors influencing churn and low savings.

Business Objective

The primary objective of this project is to analyze customer and transaction data to:

- Identify patterns in customer spending behavior
- Understand factors influencing customer churn
- Evaluate customer savings habits
- Analyze usage of digital vs traditional banking channels
- Provide actionable insights to improve retention and engagement

Key Business Questions

The analysis aims to answer the following questions:

- What percentage of customers are churning?
- How do customers spend and save money?
- Which customer segments exist based on spending behavior?
- Which segments are more likely to churn?
- Do income and age influence customer behavior?
- Which transaction channels are most used?
- Where do customers spend the most?

Data pipeline summary

Raw Data → Excel Cleaning → Pandas Transformation → SQL Analysis
→ Insights & Dashboard

Key metric defined

Total Spend: Total debit transactions per customer

Total Salary: Total salary credited

Transaction Count: Number of transactions

Savings Ratio: Salary divided by spending

Spend Segment: Customer classification based on spending